

DIGI Node — Tier 2: Complete Parts Kit

Contains every part needed to build a finished node, as delivered by the suppliers. Nothing is soldered and nothing is flashed.

In the box

Group	Contents
Electronics	Bare 82 mm PCB, all modules and headers, 8 GB microSD
Enclosure	Printed enclosure, TPU seals, all hardware and wiring
RF	WiFi and LoRa SMA antennas, U.FL pigtails and bulkheads, passive ceramic GPS antenna
Power and cooling	Fan, thermal switch, power switch, waterproof USB-C panel jack, UPS charger board, JST terminals and cables

You supply: two flat-top 18650 batteries, an active GPS SMA antenna and soldering tools.

Build it

- Follow the assembly manual from start to finish. It has photographed steps for soldering, connectors, the enclosure, the SD card and power wiring. github.com/lukeswitz/AntiHunter → **hw folder** → **Antihunter-DIGINODE-AssemblyManual.pdf**
- Two modules need their headers modified first (manual, page 6):
 - The RTC comes with female headers.
 - Some tamper sensors come with inverted 90° headers.
- Fit the SD card before you close the case. The slot cannot be reached once the PCB is mounted.

Flash it

- Flash the Heltec V3.2 or T114 with the latest stable Meshtastic from flasher.meshtastic.org, then set up the serial module as below. Without these settings the node and the radio cannot communicate. `scripts/meshtastic_config.py` in the repo applies them for you.

Setting	Heltec V3	T114
Serial module	Enabled	Enabled
Mode	TEXTMSG	TEXTMSG
Baud	115200	115200
RX / TX pins	19 / 20	10 / 9

- Flash AntiHunter on the ESP32 over USB with the web flasher at lukeswitz.github.io/AntiHunter. Choose **Full** for the web UI or **Headless** for quiet field operation.
- Set the LoRa region and change the Bluetooth pairing PIN. Make your own encrypted channel the primary channel and turn the public default off.
- Full firmware only:** join the AntiHunter WiFi AP (password antihunt3r123), open <http://192.168.4.1> and change these credentials before doing anything else.

Power

- Use a regulated 5 V supply only, from USB or the UPS board. Never wire cells or an unregulated supply to the PCB.
- A 2 A fast-blow fuse in the battery line is recommended. It is not included.

Connect one USB cable at a time. The ESP32 and the radio share the power rails.

Full guide

For unboxing through deployment, antennas, all detectors, the vibration sensor, mesh commands and Command Center, see the Operator's Guide: github.com/lukeswitz/AntiHunter/blob/main/docs/AntiHunter-Operators-Guide.pdf

- Firmware, docs, issues:** github.com/lukeswitz/AntiHunter
- Command Center:** github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO